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AI25BTECH11038 – Tejas Uppala

Question

The scalar $\mathbf{A} \cdot ((\mathbf{B} + \mathbf{C}) \times (\mathbf{A} + \mathbf{B} + \mathbf{C}))$ equals
(A) 0
(B) $[\mathbf{ABC}]$
(C) $[\mathbf{ABC}] \div [\mathbf{BCA}]$
(D) None of these

Solution

The expression is a scalar triple product, which can be written using the box product notation:

$$E = \mathbf{A} \cdot ((\mathbf{B} + \mathbf{C}) \times (\mathbf{A} + \mathbf{B} + \mathbf{C})) = [\mathbf{A} \quad (\mathbf{B} + \mathbf{C}) \quad (\mathbf{A} + \mathbf{B} + \mathbf{C})] \tag{0.1}$$

The box product is equivalent to the determinant of the matrix formed by the vectors. We can use determinant properties (where rows are the vectors $\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3$) to simplify:
Row Operation: Apply the determinant property that allows replacing one row by the sum/difference with other rows.
Operation: Replace the third vector ($\mathbf{R}_3$) by subtracting the first two vectors ($\mathbf{R}_1$ and $\mathbf{R}_2$):

$$\mathbf{R}_3' = \mathbf{R}_3 - \mathbf{R}_1 - \mathbf{R}_2 \tag{0.2}$$

The new third vector is:

$$\mathbf{R}_3' = (\mathbf{A} + \mathbf{B} + \mathbf{C}) - \mathbf{A} - (\mathbf{B} + \mathbf{C}) = \mathbf{A} + \mathbf{B} + \mathbf{C} - \mathbf{A} - \mathbf{B} - \mathbf{C0} \quad (0.3)$$

The box product becomes:

$$E = [\mathbf{A} \quad (\mathbf{B} + \mathbf{C}) \quad \mathbf{0}] \quad (0.4)$$

The determinant (or scalar triple product) is **zero** if one of the vectors is the zero vector ($\mathbf{0}$).

$$E = 0$$

The answer is **(A) 0**.